TOS

TOS

2025 10

Abstract

– *TOS* : , , , . , (AGI) . *AGI* *(2025–2500)* and *(2025–2100)*, , , / -AI . TOS , , , , .[1]

[1] , “” TOS .

Contents

1

TOS — **,** **.** AGI .
(30-day rolling, updated hourly at `metrics.tos.network`):

- /: 847 (, Sept 2025)
- : 12.4 minutes (→)
- : 1.8%
- : 99.2% (7 -)

1.1

12–24 AGI (AGIW) AI , TOS (TEM) , . 2025 8 9 2,847 97.8% 126 .[2] AGIW 83% 99.2% . 7 - 112 . , 60–80% . 2026 4 10,000 5,000 , 15 , 2% .

100 Power of AI (PAI) , / , , AGI . Bloomberg Intelligence AI 2032 $1.3 ;[3] IDC 2024 $380 2028 $1,210 .[4] McKinsey AI $2.6–$4.4 .[5] , -AI . TOS 22 AI .

1.2

, AI :

- **AI** (2028 1,210): , , AI , . TOS .
- ($850 billion by 2035): AI , (CC) (EC) , . TEM .
- ($2.5 trillion by 2045): AGI , , TOS DID , . TOS , EU .

TOS (0.1–0.5%), (12%), (8%), , . $6.1 billion 2030, 50,000 $300.

1.3

- : (DIDs), , , , RepGraph .
- **AGIW** : , .
- **/ (CC/EC)**: TEM .
- **Power of AI (PAI)**: 10,000+ TPS AI .
- : , , , .

1.4

1. : , , .
2. .
3. : , , , .

[2] TOS , "AGI Work QA Report AGIW-2025-09," September 2025.
[3] Bloomberg Intelligence, "Generative AI Market Outlook," June 2024.
[4] IDC, "Worldwide Autonomous Agent Platforms Forecast," April 2024.
[5] McKinsey Global Institute, "The economic potential of generative AI," June 2023.

2 :

2.1

“TopoSpartan” : TOS . “Topo” BlockDAG . , TOS (DAG) . . 2,500+ (TPS) , Power of AI (PAI) 10,000+ TPS .[6] DAG : . “Spartan” . , — . AGIW ; ; . , TOS : , , . : TEM (TOS) , , . “Topo” “Spartan” , — .

2.2

Economist “” AI , .[7] Forbes “ AI ” .[8] Gartner 2026 30% ,[9] Accenture 40% 3 .[10] McKinsey 1,684 , — TOS .[11]
: , AI . . , AI . Layer-2 AI . (:) , . TOS 1 .

2.3 5

AGI 10 5 , . . A .

10

MERCURY	AI , ,
VENUS	,
MARS	,
JUPITER	, - ,
SATURN	,
URANUS	,
NEPTUNE	, ,
PLUTO	,
PROXIMA	,
ANDROMEDA -	,

TOS . *MERCURY* , . *VENUS* , . *MARS JUPITER* , -AI . , *SATURN ANDROMEDA* , .
: AGI . TOS . :

AGI	( )	( )
	, DID , ,	, ,
	AGIW , ,	, ,

[6] TOS , “BlockDAG Performance Benchmark PERF-2025-08,” August 2025.
[7] *The Economist*, “Meet the agentic future,” July 13, 2024.
[8] Forbes, “Why Enterprises Need Verifiable AI Agents,” August 19, 2024.
[9] Gartner, “Top Strategic Technology Trends 2025,” October 2024.
[10] Accenture, “Autonomous Agents Are Reshaping Enterprise Automation,” August 2024.
[11] McKinsey Global Institute, “The economic potential of generative AI,” June 2023.

CC/EC , TEM , ,
, , , , -AI
PQC , , , ,

(, , ,) , , , , , , , , . : .

3

3.1

:

1: (2015–2020) (DeepMind AlphaGo, OpenAI Dota 2) (GPT-2, BERT) . . : .

2: (2020–2023) GPT-3 (2020) ChatGPT (2022) AI . : AutoGPT, LangChain, Semantic Kernel LLM . Microsoft Google "Copilot" . DeepMind AlphaCode . : .

3: (2023–) (), (SQL), ( ), ( ) AI . IDC 2024 380 2028 1,210 .[12] Accenture 40% 3 .[13] McKinsey 1,684 46% , , , — TOS .[14]
Economist "AI " .[15] Forbes " AI " .[16] : — , AI .

3.2

AI . :

, . .

AI . : (1) : 10–100 ; (2) : ; (3) *AI* : , ; (4) : . Layer-2 AI .

AI AI . AI , : (1) ; (2) ; (3) .

. ; . ( ) ( ) .

GPU/CPU . : (1) (DID,); (2) ; (3) .

3.3

Ball (: zkSNARKs) .[17] : (1) , (2) , (3) "" . TOS (BlockDAG) (AGIW) , .

[12] IDC, "Worldwide Autonomous Agent Platforms Forecast," April 2024.
[13] Accenture, "Autonomous Agents Are Reshaping Enterprise Automation," August 2024.
[14] McKinsey Global Institute, "The economic potential of generative AI," June 2023.
[15] *The Economist*, "Meet the agentic future," July 13, 2024.
[16] Forbes, "Why Enterprises Need Verifiable AI Agents," August 19, 2024.
[17] Ball et al., "Proofs of Useful Work," IACR ePrint 2017/203.

Intel SGX, AMD SEV-SNP, ARM TrustZone : . TEE AGIW , . ; TOS zk-SNARK .

(W3C DID, VC) W3C . TOS (Active Directory, OAuth) (eIDAS, GDPR) .

3.4 TOS

TOS :

- **TopoSpartan** (BlockDAG +) .
- **AGIW** (AGI) TEE .
- **CC/EC** .
- **TEM/PAI** (TOS / Power of AI) AI .
- **DID** , Web2 Web3 .
- , .

. TOS .

4

4.1

TOS 4 . .[18]

4

1. **0:** – Power of AI (PAI) BlockDAG .
2. **1:** – (DID), (VC), (RepGraph).
3. **2:** – AGI (AGIW) , / (CC/EC), TOS (TEM).
4. **3:** – , , , .

- : , , .
- : API, , AML/KYC .
- : 24/7 SOC ( ), , .
- : SDK(Python, JavaScript, Rust), CLI , , .

4.2 0:

BlockDAG TOS 1–3 (DAG) . : . DAG (2,500+ TPS, PAI 10,000+ TPS) .[19] ; .

libp2p(, NAT ,) . . . (, ,) gRPC .

, Merkle (Patricia Merkle Trie) . .

[18] `explorer.tos.network` .
[19] TOS , "BlockDAG Performance Benchmark PERF-2025-08," August 2025.

4.3 1:

DID W3C (DID) . DID , (: , URI) . `registerDID(didDocument)` . DID . ; 0 .

(VC) (, ,) DID VC . VC (IPFS, Arweave) . : “ , “ , “1000 . .

Merkle : ID Merkle . (Merkle) . .

RepGraph ( ) , , $r_a \in [0, 1]$. **??** . , . , (v4.0) ( ) .

Pedersen Bulletproof . ( ). .

4.4 2: AGIW

Task Factory TaskInstance . (, , , ,) . , (: 0.01 TOS) , . .

Task Instance . : `Open` → `Claimed` → `Submitted` → `Validating` → `Settled` (`Disputed`). `claimTask()` ( ), `submitWork(resultHash, attestation)` . .

TOS . 3–7 ( ). (: , ,) `validateWork(taskID, score)` . ; (>2) . : $Acc_{t+1} = 0.9 \times Acc_t + 0.1 \times correctness_t$.

$q \in [0, 1]$: $e \times q$, $e \times (1 - q) \times \beta$, $e \times \gamma$. : $\beta = 0.12$, $\gamma = 0.08$. . (5–60) .

. . ( ) , , . ; . (TEM) .

4.5 3:

CC/EC (CC) TFLOP ; (EC) kWh . AWS, Azure, GCP(CC) ISO (EC) . CC/EC TOS , .

21 ( ) TOS 15 . . (30) . (±5%) .

TEM TOS , . , 2% TEM . CC/EC (: 20%) TEM (AMM) . TEM .

(AMM) CC/EC ↔ TOS . (0.3%) ; TEM . (Uniswap v3) .

4.6 4:

(DSL) . : “10,000 TOS .” (, ,) Python . , , .

(: “ ,” “ 3/5 ”). (75%) . “” , .

(7–15). ( ), (,) . .

(, ,) . , . API (: AGIW ,) .

4.7

(IDS) gRPC , Apache Arrow (,) , GraphQL API . (: ,) ; . IDS TOS , .

API AGIW , . : “ ,” “ ,” “CC/EC .” , OAuth2 , .

SDK Rust, Python, TypeScript . : `tos-sdk-py` `Agent.register_did()`, `Task.publish()`, `Task.claim()`, `Task.submit()` . Terraform . GitHub Actions CI/CD .

4.8 ()

:

1. ( )
   a. , DID , `DIDRegistry.register()` .
   b. KYC , `KYC-Verified` .
   c. TOS(: 100 TOS) ; 7 .
2. ( )
   a. (: ,).
   b. ; .
3. ( )
   a. TaskFactory .
   b. `TaskInstance.claim()` ($\geq$,).
   c. `Claimed` ; ; .
4. (+)
   a. (TEE) : Intel SGX, AMD SEV-SNP ARM Realm.
   b. TEE : ( ), / , , .
   c. , , `submitWork()` .
5. ( )
   a. ValidationPool 3–7 (,).
   b. , , .
   c. (), , .
6. (+)
   a. RewardSplitter : CC/EC $e \times q$, , .
   b. RepGraph : $r_{t+1} = 0.9 \times r_t + 0.1 \times q$.
   c. ; ArbitrationContract .
7. ( )
   a. (+).
   b. (: PQC).
   c. , .

4.9 :

1. ( ) : “500 PII .” : 50 CC () + 5 EC (). : 2.
2. **A** (DID `did:tos:tos1qpz8vq6qgnkgw9zzq9z5qgpgqx8qgz5dpgrxve`, 0.87, AI) . : `Claimed`.
3. **A** Intel SGX . , (4.8 EC), (22) .
4. **A** ( ) + . : `Submitted`.
5. (3): , 10% , . : 0.95, 0.94, 0.96. : 0.95.
6. **RewardSplitter** A $50 \times 0.95 = 47.5$ CC + $5 \times 0.95 = 4.75$ EC . 6 CC + 0.6 EC . 4 CC + 0.4 EC .
7. **RepGraph** A : $r_{t+1} = 0.9 \times 0.87 + 0.1 \times 0.95 = 0.878$.
8. , , . (GDPR 30).

AI .

5 ( ) –

5.1

. : 30% , $120 .[20] , . , , .

5.2 12–18

D1. v1

- BLS DID , DID .
- , (VC) ; Merkle .
- Pedersen Bulletproof ; .

D2. AGI (AGIW) v1

- Task Factory Task Instance , , .
- Intel SGX, AMD SEV-SNP ARM Realm ; .
- Validation Pool ; .
- Reward Splitter CC/EC , .

D3. /

- TFLOP CC ; kWh EC . .
- Paymaster CC/EC ; TEM (:) .

D4.

- , , AGIW , .
- , , , , .
- Rust, Python, TypeScript SDK; Terraform .

[20] Gartner, “Top Strategic Technology Trends 2025,” October 2024; IDC, “Worldwide Autonomous Agent Platforms Forecast,” April 2024.

5.3

	(12)	
	5,000	DID , , paymaster
	10,000	AGIW , ,
	< 15	,
	< 2%	
CC/EC	> 60%	TEM (,)
	60–80%	
	SOC 2	API ,

5.4 ( )

DID DID Python (`DIDRegistry.py`):

```
class DIDRegistry:
    @public
    def register_did(self, did_hash: bytes, did_document: bytes,
                     signature: bytes) -> None:
        """ DID  """

    @public
    def update_did(self, did_hash: bytes, new_document: bytes,
                   signature: bytes) -> None:
        """ DID  """

    @public
    def revoke_did(self, did_hash: bytes, signature: bytes) -> None:
        """DID """

    @public
    @view
    def resolve_did(self, did_hash: bytes) -> bytes:
        """DID  """
```

DID W3C DID Core (v1.0) . :

- `id`: (: `did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xhyepcg...`)
- `verificationMethod`:
- `service`: , , API
- `created`, `updated`:

: 80,000 ($2,000 TOS 50 gwei $0.50), .

W3C VC (v1.1) . JSON-LD :

```
{
  "@context": ["https://www.w3.org/2018/credentials/v1"],
  "id": "https://issuer.example/credentials/987",
```

```
  "type": ["VerifiableCredential", "SkillCertificate"],
  "issuer": "did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh",
  "issuanceDate": "2025-10-01T00:00:00Z",
  "credentialSubject": {
    "id": "did:tos:tos1qq5hqy2jwn0r5ehvuew5r3qjcfvjvq5dhldxvyq",
    "skill": " ",
    "proficiencyLevel": ""
  },
  "proof": {
    "type": "BbsBlsSignature2020",
    "created": "2025-10-01T00:00:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh#key-1",
    "proofValue": "..."
  }
}
```


: VC (IPFS, Arweave) ; (32) . VC , , .

2^{256} Merkle . ; ID . ID Merkle ( ). . : 50,000 ($0.30).

Pedersen :

$$C(v, r) = vG + rH \tag{1}$$

v (), r , G H . : $C(v_1, r_1) + C(v_2, r_2) = C(v_1 + v_2, r_1 + r_2)$, . Bulletproof $v \in [0, 2^{64})$. : 672, . : 1.2ms. ; `enableConfidentialMode()` .

5.5 AGIW ( )

AGIW :

1. TaskFactory (`TaskFactory.py`)
: .
:

- `publishTask(metadata, escrow, deadline, allowedAgents)`: TaskInstance , , 0.01 TOS , `TaskPublished` . ID .
- `listTasks(filters)`: ; , , .
- `updateWhitelist(agentDIDs)`: (: KYC , > 0.7) .

:

- : 0.01 TOS(,).
- : .
- : 150,000 (50 gwei $0.90).

: ; `RewardSplitter` .

2. TaskInstance (`TaskInstance.py`)

: .

:

$$\texttt{Open} \xrightarrow{\text{claim()}} \texttt{Claimed} \xrightarrow{\text{submitWork()}} \texttt{Submitted} \xrightarrow{\text{validate()}} \texttt{Validating} \xrightarrow{\text{consensus}} \texttt{Settled} \tag{2}$$

: $\xrightarrow{\text{dispute()}}$ `Disputed` $\xrightarrow{\text{arbitration}}$ `Resolved`.

:

```
@dataclass
class Task:
    task_id: bytes              #
    publisher: str              #
    claimed_by: str             #
    escrow: int                 #
    deadline: int               #
    spec_hash: bytes            #  IPFS CID
    result_hash: bytes          #
    attestation: bytes          # TEE
    state: TaskState            # : Open, Claimed, Submitted, ...
    validator_scores: dict[str, int]  #  [0,100]
    final_quality: int          #
```

:

- `claimTask(agentDID)`: (1) `Open` , (2) , (3) $\geq$. `Claimed` , .
- `submitWork(resultHash, attestation)`: (1) , (2) , (3) . `Submitted` , `WorkSubmitted` .
- `validateWork(score)`: . $\in [0, 100]$. , `Settled` , `RewardSplitter` .

:

- `forfeit()` , 10% ( ) . 0.05 .
- 24 : ; 5% .

3. ValidationPool (`ValidationPool.py`)

: , , .

:

- 1,000 TOS .
- Acc_v 0.5 , :

$$Acc_{v,t+1} = \alpha \times Acc_{v,t} + (1 - \alpha) \times correctness_t, \quad \alpha = 0.9 \tag{3}$$

$correctness_t \in \{0, 1\}$ ($\pm 10\%$) .

:

1. `WorkSubmitted` , ValidationPool $n = 5$ ( ).
2. :

$$P(\text{select } v) \propto stake_v \times (1 + Acc_v) \tag{4}$$

3. ; 24 .

:

- : $> 2\sigma$ 1
- : 5
- : 3 > 10 ( ), ; 20

: $e \times \beta \times (1 - q)$. e , $\beta = 0.12$, q . Acc_v .

4. RewardSplitter (`RewardSplitter.py`)

: .

: e (CC/EC) $q \in [0, 1]$:

$$R_{\text{agent}} = e \times q \times (1 + \alpha(r_a - 0.5)) \times f(s_a) \tag{5}$$

$$R_{\text{validators}} = e \times \beta \times (1 - q) \tag{6}$$

$$R_{\text{treasury}} = e \times \gamma + \text{slashed funds} \tag{7}$$

:

- $r_a \in [0, 1]$ (**??**),
- $\alpha = 0.2$,
- $f(s_a) = \min(2, 1 + \log(1 + s_a))$ ( ),
- $\beta = 0.12$,
- $\gamma = 0.08$.

: (**??**):

- $(r \geq 0.90)$: 5.
- $(0.70 \leq r < 0.90)$: 15.
- $(0.30 \leq r < 0.70)$: 30.
- $(r < 0.30)$: 60.

.

: RewardSplitter CC/EC ~30,000 ($0.18) .

5.6

CC/EC 21 . TOS API (, ISO) . . . .

5.7

:

- Apache Arrow gRPC .
- .
- , , (Grafana).
- API .

5.8

T , A , V . t c_t e_t . U_a :

$$U_a = \sum_{t \in T_a} q_{a,t} \cdot e_t - \gamma s_a - p_a \tag{8}$$

$q_{a,t}$, s_a , p_a , γ . U_v . :

$$\sum_{a \in A} s_a = \theta \sum_{t \in T} e_t, \quad 0.2 \leq \theta \leq 0.4 \tag{9}$$

.

:

$$Reward_a(t) = e_t \times q_{a,t} \times (1 + \alpha(r_a - 0.5)) \times f(s_a) \tag{10}$$

$f(s_a) = \min(2, 1 + \log(1 + s_a))$ r_a . $Reward_v(t) = e_t \times \beta \times Acc_v(t)$, Acc_v . δs .

5.9 (RepGraph)

RepGraph (, ,) (,) .

a r_a^{base} :

$$r_a^{\text{base}} = 0.3 \times \frac{\text{age}_a}{\text{age}_{\max}} + 0.4 \times \frac{\text{txCount}_a}{\text{txCount}_{\max}} + 0.3 \times \frac{\text{stake}_a}{\text{stake}_{\max}} \tag{11}$$

:

- age_a (365),
- txCount_a ,
- stake_a TOS ,
- .

q_t , Acc_a :

$$Acc_{a,t+1} = 0.9 \times Acc_{a,t} + 0.1 \times q_t \tag{12}$$

± 0.2 :

$$r_a^{\text{accuracy}} = r_a^{\text{base}} + 0.2 \times (2Acc_a - 1) \tag{13}$$

$(Acc_a \to 1 \Rightarrow +0.2)$ $(Acc_a \to 0 \Rightarrow -0.2)$.

> 100 $Acc_a > 0.85$ $+0.1$, $r_a = 1.0$.

r_a :

Platinum	$r \geq 0.90$	5	30–50%	14%
Gold	$0.70 \leq r < 0.90$	15	10–30%	33%
Silver	$0.30 \leq r < 0.70$	30	0–10%	41%
Bronze	$r < 0.30$	60	: 2×	12%

Table 4: 2025 8 RepGraph .

RepGraph 7 :

1. : 7 100 TOS ( ).
2. : (< 7) (< 10 CC) .
3. **IP** : IP ; .
4. : .
5. : (KYC,) .
6. (v4.0): BrightID, Worldcoin Gitcoin Passport .
7. : RepGraph ; .

2025 8 , 0.7% 112 .[21]

5.10

(SLO) Foundations :

- **SLO1** – : AGIW 99.9% 20 (p99.9 < 20).
- **SLO2** – : CC/EC 15 ; (> 30) .
- **SLO3** – : < 2%; TEM 20% .
- **SLO4** – : ≥ 99.95%(< 4.4).
- **SLO5** – : ( ).

:

- : (//, ,) gRPC (IDS) .
- : IDS Apache Arrow (10× ,).
- : (, ,) SHA-256 ; .
- : Grafana , , , , .
- **API**: RESTful (OAuth2 ,) : " AGIW ," " ," "CC/EC ."

Prometheus SLO (: TEM ,) .

5.11

(∈ [0, 1], ∈ [100, 10000] TOS) 10,000 :

- : 30 ±0.05 (95%).
- : ($r < 0.3$) 50 Gold ($r > 0.7$) .
- : 42% ; (5-60) 6% .
- : 10% 28% (0.34) ; α( ) $f(s_a)$( ) .

5.12

2025 8-9 devnet(Aurora) testnet(Helios) AGIW .

[20]TOS , "Reputation & Anti-Sybil Telemetry REP-2025-08," August 2025.
[21]TOS , "Reputation & Anti-Sybil Telemetry REP-2025-08," August 2025.

(2025 8 15 – 9 15)

- : 2,847 (, ,).
- : 126(, ,).
- : 97.8%(126 123 ; 3 ,).
- : 83%(: 12.3).
- : 99.2% (112 , 1).
- : 60–80% .

(1,000 , 2025 9)

- : 7.4( ).
- : 3.1%(TEE + 230ms).
- : 0.002 TOS(+).
- : 45 (2,700).
- : 50 (,) ; 100% .

5.13

TOS Foundations AI :

- **US AI RMF**(NIST): , , (, ,) .
- **EU AI Act**: AI (,) KYC, TOS .
- **Singapore MAS** : AI ; TOS API MAS .

AI TOS :

- : AI (12%).
- : , ( ).
- : / ( ).
- : ( ).

(**??**) 10 TOS 3-4 .

EC . ' ' ; / 10% (TEM). .

6 – MERCURY VENUS

6.1

MERCURY VENUS AI , .[22] , .

6.2

Markets & Safety :

[22] McKinsey Global Institute, "The economic potential of generative AI," June 2023.

1. (`CustodyVault.py`)
: , . .
:

- : m (, ,) T m-of-n . : $> 10,000$ TOS 2-of-3 (+ CFO, + , CFO +).
- **(HSMs)**: FIPS 140-2 Level 3 HSM(AWS CloudHSM, Thales Luna) . HSM ; .
- : :
  - : // $X.
  - : (: ,).
  - : (OFAC).
  - : .
- : .

: SOC 2 Type II HSM , , . SAML/OAuth (Active Directory, Okta) .

2. (`BridgeController.py`, `FiatGateway.py`)
: (, ,) .
:

- **HTLC** : TOS . : TOS H 100 TOS ; H ; , 24 .
- : (7-of-11) ; ($< \$1M$) . (TLA+) .

:

- **API**: ISO20022 (SWIFT, FedNow, SEPA) . USD/EUR ; CC/EC .
- : Chainalysis, Elliptic KYC/AML ; (>95%) .
- : 2-4 ( ); 24-48( ).

:

- : CC (GPU). : 20%, 10%.
- : EC ; / .
- : ; (3%) .

3. (`ArbitrationCourt.py`)
: (,) .
:

- : (, AI ,) , TOS , 10,000 TOS .
- : (, , ,) ; .
- : ; (20 $< 3.5/5$) .

:

1. : ( ) (5-10%) .
2. : 3 (, ,).
3. : (, ,) IPFS ; .
4. : , , .
5. : (2-of-3) (, ,). .
6. : ; (+).

: TEM (AGIW 2%) ; (:) . 4-of-7 .

4. (`SafetyOracle.py`)
: AI (, ,) ; .
:

- **AI** : Anthropic Constitutional AI , OpenAI GPT , Google DeepMind .[23]
- **API**: Perspective API(), AWS Comprehend(, PII), Microsoft Azure Content Safety.
- : , , (GDPR 6) .

:

- : Safety Oracle (: > 0.9, PII) ; .
- : (30 >3) (: 10) .
- : 'GDPR ' ; .

: Safety Oracle : , , ($< 2\%$).

6.3

x . . Safety Oracle ; .

6.4

HTLC . ISO20022 . , USD ; CC . CC/EC .

6.5

?? .

,	( )
,	()
HSM, ,	( )
,	( )
,	( )

Table 5: - .

6.6 Power of AI (PAI)

PAI TOS BlockDAG (2,500+ TPS) 10,000+ TPS AI .

1: AI Scheduling Hints (MERCURY) : AI ( ) . (, AGIW ,) . $> 67\%$ AI ; BlockDAG .
: ; . : AI $\Rightarrow$ BlockDAG .
: 20-30% .
: AI (Apache 2.0) , , ( ).

[23] IBM Research, "AI Safety Metrics for Enterprise Deployment," May 2024.

2: (MERCURY) : :

- **AI** (30%): AI , AGIW .
- (70%): AI , (: ,) , .

: k $1 - 2^{-k}$ ' '. AI ; .
: TLA+ ; (< 33%) ( ) ( ) .
: 1 3-4× (7,500-10,000 TPS).
: AI (: ,) . 4-of-7 .

3: Full PAI Consensus (MERCURY / VENUS) : AI :

- : , (DDoS,) .
- : DAG() ; AI .
- : AI , TEM .

: 10,000+ TPS, 15,000+ TPS, 95% 500ms .
: :

- : 500( ).
- : < 0.45( ).
- AI : , , .

: PAI (**??**), (/) (, AGI) .

6.7

(MERCURY)
AGIW , (HSM,), / , AI . (60-80%) (SOC 2, EU AI Act) .

(VENUS ANDROMEDA)
, (DAE) . PAI , -AI , .

6.8

	, (1,000 TOS), EWMA , , , RepGraph	(11/21 ;) (;)
( )	HSM (FIPS 140-2 3), , , , ( ), (ML),	(HSM +) (; TEM)
( )	(Anthropic, OpenAI, Google,), ,	(;)
PAI	AI , (,), , (TLA+), (7-of-11), ($1M/tx), (3%)	(;) (4+ +)

Table 6: - (MERCURY-VENUS).

7 – VENUS MARS

7.1 -

2070 Metaworld 1.0 . AI . , , .

7.2 -

Motivation , . . .

1. **Policy DSL (Domain-Specific Language)**: . :

```
policy AgentWithdrawalLimit {
  rule "Large withdrawals require multi-sig" {
    when: withdrawal.amount > 10000 TOS
    require: signatures.count >= 2
    from: [agent.key, guardian.key, auditor.key]
    timeout: 24 hours
  }
  rule "Sanctioned addresses blocked" {
    when: transfer.recipient in SanctionsList
    action: reject
    log: "OFAC violation attempt"
  }
}
```

2. :
   - : DSL (AST) .
   - : ( ), ( ), .
   - : TOS Python .
   - : , .
   - : SMT (Z3, CVC5) (: ' ', ' ') .
3. :
   - DSL GitHub .
   - CI/CD : , , .
   - , , (7).
   - : 60% , 75% .
   - : 48 ( ).

:

- : ( ).
- : (`onlyAgent`, `onlyGuardian`, `only`) .
- : , ID .
- : ; ; .

- : ' X Y '(ITAR).
- : ' 1,000 EC ; '
- : 'Safety Oracle 1 > 10 '

7.3

CRDT . **??** .

Algorithm 1 (DTS)

: i L_i.
$d_i = H(L_i)$.
d_j .
$L_i \leftarrow \text{merge}(L_i, L_j)$.
.

7.4

SLO: 75% , ≥ 48, . , , .

7.5

- , .
- -AI , .
- .

8 & – JUPITER ANDROMEDA

AGI (JUPITER) (ANDROMEDA), TOS , . TOS - .

8.1

: - 8 44 . 2.5 . (,) 40-80 . (BFT, PoS) 1 ; :

- : .
- : ( ).
- : .

: TOS 2 :

1: (LCDs)
(, , ,) TOS . LCD BlockDAG + PAI (1). ; .

2: (IRC)
24( ) LCD IRC Merkle zk-SNARK . IRC :

1. **BFT**: Tendermint; 60 .
2. : 51% (67%), .
3. : 7( ) IRC LCD .

. :

1. H 48 LCD HTLC( ) .
2. LCD IRC HTLC ( ).
3. LCD IRC , HTLC , x ($H = \text{hash}(x)$).
4. LCD IRC .
5. LCD , ( ).

: = 48-96(15). .

: 40% , LCD IRC .
: IRC Merkle , , . LCD : , . : 50% ; LCD .

(JUPITER)

- **TPS**: LCD 10,000+(PAI).
- : 24-48( ).
- : 7( ).
- : 30 ( ).

8.2

: AGI (, ,) . :

- ($50B+).
- .
- ( ) .

. TOS .

:
24 LCD :

- : , , Merkle .
- : zk-SNARK(, ,).
- : , , , .

IRC ( ).

(, 75%):

1. : (, ,) .
2. : (,) *(RAC)* :
   - ID(,).
   - (,).
   - (, : 3).
3. : RAC ; 75% + .
4. : . ; .

- : ; (' ') . .
- : 90 ( ). .
- : 5% (:).
- : 1 ; .

: (2185) : 100 ECDSA 8 5,000 120 (PQC).
:

1. (10,000).
2. ; 2 .
3. ; .
4. (82%) 12 .
5. ; ; .
6. 2 .
7. : PQC ; 30 Dilithium-5 ; .

: 7 ; ; .

8.3

: AGI . :

- 200 .
- ( ).
- (/AI).
- (X 2234 Y ?).

(AWS Glacier, Google Coldline) <100 ; , , .

: 1: (1–10)
SSD/NVMe + . . : LCD 3 , (: 3 ; : Olympus Mons, Valles Marineris, Hellas Basin).
2: (10–100)
HDD . Reed-Solomon (20%) 20% . ; . : TB 0.5 TOS .
3: (100–1,000)
:

- **DNA** : DNA (Church Lab, Harvard). : 215; : -18řC 2,000 . : TB $1,000(2025 ; 2050 TB $10).
- **5D** : (Southampton). : 138 ( ). : 360TB.
- **(LTO-12+)**: . : 50-100( ).

: (7) ; (: Svalbard , ; : ; :).
4:
:

1. ; ; ( ).
2. : Merkle .
3. (:) .

(: '2234-05-15 tos1qq5hqy2jwn0r5ehvuew5r3qjcfvjvq5dhldxvyq ?'):

1. Hot Storage (10): O(log n) , 100ms .
2. Warm Storage (10-100): , Merkle , 10-60.
3. Cold Storage (100-1,000): DNA/ , /, Reed-Solomon , 1-7.

: 3% . 500 (2150) 15 :

- DNA (5 , 500 PB).
- 5D (3 , 1,500 PB).
- /(4).
- (3 , 1,000).

8.4

- **2030**: 50 (ECDSA).
- **2040**: 1,000 (Shor RSA-2048, ECDSA-256).
- **2050**: .

: 1 (v4.0, 2026): . :

- : ECDSA secp256k1( ).
- : Dilithium-3 (NIST FIPS 204).

; (PQC).
2 (v5.0, 2030): PQC . ; ECDSA . : 12.
3 (v6.0, 2040): . SHA-256(Grover) SHA-3 BLAKE3 (256).

(&) TOS *PQC* :

1. **1 (1–3)**: (, ,). 5,000 PQC .
2. **2 (4)**: (MPC) . ; .
3. **3 (5–6)**: . PQC ; .
4. **4 (7)**: (67%). 30 .
5. **5 (8)**: . 12 .

PQC (:):

- PQC .
- PQC (Falcon, SPHINCS+).
- .

8.5 :

: . ; TOS . TOS AGI :

- (,).
- (, ,).
- (,).

(90 , 75% ,) + . AGI .

9

TOS . .

9.1 P1 –

(DIDs) , `did:tos:` W3C DID( ) .
DID :

```
did:tos:<bech32-address>
did:tos:<network-id>:<bech32-address>
```

:

: `did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh`
: `did:tos:mainnet:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh`
: `did:tos:testnet:tst1qqqsyqcyq5rqwzqfpg9scrgwpugpzysnzs2pgde5k`

: TOS bech32 DID , . `tos1` , `tst1` .
DID :

- : Ed25519(), X25519(), Dilithium-3( ).
- : HTTPS URL, IPFS .
- : (, ,) .
- : .

:

1. DID .
2. TOS (: 1 TOS).
3. DID , DID IPFS , .
4. .

(VCs) (, ,) .
:

- : RepGraph (Bronze/Silver/Gold/Platinum), , .
- : (, ,).
- : KYC/AML ( ), (EU AI Act , FDA).
- : (DAO).

:

1. (: RepGraph) JSON-LD .
2. BBS+ (;).
3. IPFS DID .
4. (: “ ≥ 0.7 ”).

(RepGraph) :

$$r_t = \alpha \cdot r_{t-1} + (1 - \alpha) \cdot s_t \tag{14}$$

:

- r_t: t ([0, 1]).
- $\alpha = 0.85$: (85%).
- s_t: (5).

:

$$r_{t+\Delta t} = r_t \cdot (0.98)^{\Delta t/30} \tag{15}$$

2% . .

:

- : .
- : , , , .
- : (30) 2× .

:

- : 100 TOS + (BrightID v5.0).
- : .
- : 90 0.5 , .

W3C

- **DID Core v1.0**: ; TOS DID Universal Resolver .
- **v1.1**: (TOS TOS).
- **DIDComm v2**: (, ,).

9.2 P2 –

TOS . AI .

:

- : 3 (:).
- : m-of-n ( ).
- : ( ).
- : (,) .

UserOperation :

```
{
  sender: "0x...",           //
  nonce: 42,                 //
  initCode: "0x...",         //    ( )
  callData: "0x...",         //
  paymasterAndData: "0x...", // Paymaster  +
  signature: "0x..."         // UserOperation
}
```

AGIW . Section **??** .

:

- **TaskInstance** : , (Open → Claimed → Verified → Settled).
- : 5 ; .
- : TEE (Intel SGX, AMD SEV).
- : .

(CC/EC) **(CC)**: GPU/CPU . AWS/GCP/Azure .
(EC): (kWh). ; 10% .
(AMM):

- (CPMM): $x \cdot y = k$ $x =$ CC , $y =$ TOS .
- 0.3% .
- : TEM 1M TOS LP .

TOS **(TEM)** .
:

$$= \quad + \quad - \text{TEM} \tag{16}$$

:

- .
- : 50% ; 12.5% .

TEM :

- **(100%)**: , , .
- **(75%)**: , .
- **(50%)**: <$10M .
- **(0%)**: .

: 8% TEM . .

Table 7: TEM

	/	
(TOS)	0.0001–0.01	p95 ; 50%
	+12.5%	50%
CC	GPU ±15%	30
EC	kWh ±10%	
TEM	40% / 30% / 30%	
	≤30	CC/EC
	: 100% / : 75% / SMB: 50%	
	/ : 0%	
	8%	
	60%	
	10%	

: TEM . (`metrics.tos.network/tem-sensitivity`) 3× 50% .

9.3 P3 –

BlockDAG , TOS (DAG) .
:

- 1-3 (1).
- .
- , + .

:

- : 8 (90).
- : DAG .
- : (24) 100 , + .

Performance:

- : 2,500-3,500 TPS(2025 9).
- PAI : 10,000+ TPS(2026 4).

Table 8:

Mempool	8 (, , -, , , , ,) 32 ; BLS 16 vCPU, 64 GB RAM, NVMe SSD (1 TB), 2.5 Gbps RTT: 20–180ms ( ), : 0.1–1.0%, : ±10ms : (15%), (25%), RAG (20%); : 20–60% 72 ; 5 ; + : 100–2,000 ; 50% ; 100,000 TPS (p50/p95/p99), (p50/p95), ( ), AGIW (→), (%), (%), Prometheus + Grafana; ; `metrics.tos.network` 2025 9 (14)

(2025 9):

- TPS: p50 = 2,847, p95 = 3,421, p99 = 3,789
- Block time: p50 = 11.3s, p95 = 14.2s
- : 8 (90)
- AGIW : p50 = 12.4, p95 = 18.7
- : 1.8%
- : 98.7%

Power of AI (PAI) AI BlockDAG .

1: AI Scheduling Hints

ML (LSTM) . AI , >67% .

2: Hybrid Committees

AI (30%) , (70%) . : k $1 - 2^{-k}$.

3: Full PAI

AI . AI 40% (AI), .

AI AI :

- **Anthropic Constitutional AI**: (, ,).
- **OpenAI Moderation API**: (, ,).
- **Google Perspective**: .

Enforcement:

- < 0.3 ( ).
- .
- (0.3-0.5) .

- :

- : $\Rightarrow$ 50% .
- : 6 $\Rightarrow$ 5% , .
- : TEE $\Rightarrow$ 20% .

:

- : ID 100 TOS.
- : 5-60 ( ).
- : .

9.4 P4 –

PQC Section **??** . :

- **v4.0 (2026)**: (ECDSA + Dilithium-3).
- **v5.0 (2030)**: PQC ; .
- **v6.0 (2040)**: (SHA-3).

. 24 ; 90 ; 75% . Section **??** .

DNA 5D (Hot/Warm/Cold). . Section **??** .

(+) . 48-96 . Section **??** .

:

- (-): LCD .
- (<33%): 5 .
- (>33%): ; + .

SLA:

- (1-5%): 1 + .
- (5-20%): ; 24 .
- (>20%): ; ; 7 .

10 AGI (AGIW)

10.1

T :

$$T = (id, owner, spec, escrow_{CC}, escrow_{EC}, stake_{req}, deadline, whitelist, state) \tag{17}$$

:

- $id \in \{0,1\}^{256}$: (keccak256).
- $owner \in \mathbb{A}$: .
- $spec$: IPFS ID(JSON , ,).
- $escrow_{CC}, escrow_{EC} \in \mathbb{R}^+$: .
- $stake_{req} \in \mathbb{R}^+$: .
- $deadline \in \mathbb{N}$: .
- $whitelist \subseteq \mathbb{A}$: ($\Rightarrow$).
- $state \in \{\texttt{Open}, \texttt{Claimed}, \texttt{Submitted}, \texttt{Validating}, \texttt{Settled}, \texttt{Disputed}\}$.

a :

$$W = (resultHash, attestation, metrics) \tag{18}$$

:

- $resultHash \in \{0,1\}^{256}$: (IPFS CID).
- $attestation$: TEE :
  - $measurementHash$: SHA-256 ( ).
  - $inputHash, outputHash$: .
  - $timestamp$: / .
  - $energyUsed \in \mathbb{R}^+$: kWh(OS TEE).
  - $signature$: TEE (Intel SGX: secp256r1 ECDSA; AMD SEV: RSA-4096).
- $metrics$: (, ,).

:

$$\mathcal{V} : (W, spec, T) \to [0,1] \tag{19}$$

:

1. : (Intel, AMD) TEE .
2. : $measurementHash$.
3. : $resultHash$ ( ) ( ) .
4. : $energyUsed \leq spec$.

$q_v \in [0,1]$. :

$$q = \text{median}(\{q_v : v \in \text{assigned validators}\}) \tag{20}$$

$\geq 67\%$ $|q_v - q| \leq \epsilon$, $\epsilon = 0.1$.

$> 2\sigma$:

$$slash_v = \min(0.05 \times stake_v,\ 0.01 \times stake_v \times |q_v - q|/\epsilon) \tag{21}$$

. (100 > 5) .

10.2

AGIW :

`Open`	`claimTask()`	*whitelist* ( ), stake $\geq$ $stake_{req}$, block $<$ deadline	`Claimed`
`Claimed`	`submitWork()`	, block $<$ deadline,	`Submitted`
`Claimed`	Block $\geq$ deadline		`Forfeited` (10%)
`Submitted`		( )	`Validating`
`Validating`		ϵ $\geq 67\%$	`Settled`
`Validating`			`Disputed`
`Settled`		(RewardSplitter)	
`Disputed`		(2-of-3)	`Resolved`
`Resolved`			

Table 9: , AGIW .

10.3 ()

1:

1. (TEE) : Intel SGX , AMD SEV-SNP VM ARM Realm.
2. ( ), TEE .
3. TEE : + + .

2: TEE

1. TEE (OS).
2. TEE :
   - **Intel SGX**: RAPL(Running Average Power Limit) .
   - **AMD SEV**: SMU(System Management Unit) .
   - **ARM Realm**: PMU(Performance Monitoring Unit) .
3. TEE $resultHash = \texttt{keccak256}(output)$.

3: TEE :

```
{
  "version": "2.0",
  "timestamp": 1698765432,
  "measurementHash": "sha256:abcd...1234",
  "inputHash": "sha256:5678...ef90",
  "outputHash": "sha256:1111...2222",
  "energyUsed_kWh": 4.8,
  "runtime_seconds": 1320,
  "teeType": "IntelSGX",
  "signature": "base64:M3Mz...NDQ0"
}
```

TEE (,) .

4: `TaskInstance.submitWork(resultHash, attestation)` . :

- TEE ( ) *signature* .
- `WorkSubmitted(taskID, agentDID, resultHash, attestation)` .
- ValidationPool .

5: :

1. .
2. TEE ( ).
3. TEE $resultHash$.
4. $energyUsed \leq$.
5. `ValidationPool.validateWork(taskID, q_v)` $q_v \in [0, 1]$.

: zk-SNARK (2027+) TEE :

- x, y " x C y " zk-SNARK .
- : ∼200( ).
- : ∼5ms on commodity hardware (vs. TEE signature verification ∼2ms).
- : (Intel, AMD) ; .

11 /

11.1 Oracles

21 . . .

11.2

CC/EC . TEM . .

12

12.1 Methodology

100,000 () TOS . :

- : t $N_a(t)$.
- : $V_t(t)$.
- : $\bar{e}_t$(USD CC/EC).
- : $\bar{q}_t$, σ_q.
- : $d_t \in [0, 1]$ fraction of tasks escalating to arbitration.
- : t $N_v(t)$.

:

- : . ($V_t \propto N_a^{0.85}$).
- : RepGraph EMA . .
- : TEM , .

12.2

(N_a)	20,000	50,000	150,000
/ (3)	50,000	150,000	500,000
($)	$150	$300	$600
$\bar{q}$	0.75	0.85	0.92
d	4%	2%	0.8%
(N_v)	200	500	1,200
TOS (3)	$2	$5	$15
	(5)	(3)	(2)

Table 10: TOS (2025–2030).

12.3 Formulas

y $S(y)$:

$$S(y) = 365 \times V_t(y) \times \bar{e}_t(y) \tag{22}$$

(, 3): $S(3) = 365 \times 150{,}000 \times \$300 = \$16.4$.

:

$$I_a(y) = \frac{V_t(y)}{N_a(y)} \times \bar{e}_t(y) \times \bar{q}(y) \times (1 + 0.2(r_a - 0.5)) \times f(s_a) \tag{23}$$

($r_a = 0.6$, $s_a = 500$ TOS):

$$I_a(3) = \frac{150{,}000}{50{,}000} \times \$300 \times 0.85 \times 1.02 \times 1.62 \approx \$1{,}250 \tag{24}$$

(Platinum, $r_a = 0.95$, $s_a = 5{,}000$ TOS):

$$I_a^{\text{top}}(3) \approx \$8{,}200 \tag{25}$$

(12%,):

$$I_v(y) = \frac{0.12 \times S(y)}{N_v(y)} \times \frac{Acc_v}{\bar{Acc}} \tag{26}$$

3, ($Acc_v = 0.85$):

$$I_v(3) = \frac{0.12 \times \$16.4B}{500} \times \frac{0.85}{0.85} \approx \$3.9M \tag{27}$$

8% + :

$$T(y) = 0.08 \times S(y) + \sum_{slash\ events} slash_v \tag{28}$$

3: $T(3) = 0.08 \times \$16.4B + \$20M \approx \$1.33B$.

12.4

(,)

- 3 : 365 × 50,000 × $150 = $2.74 .
- : $410/; $2,800/.
- : $1.3M/().
- : $240M.
- : . . .

(,)

- 3 : $16.4 .
- : $1,250/; $8,200/.
- : $3.9M/().
- : $1.33B.
- : ; . . PAI , , .

(,)

- 3 : 365 × 500,000 × $600 = $109.5 .
- : $2,190/; $16,800/.
- : $10.9M/().
- : $8.8B.
- : TOS . . . (PAI 3, , DAE) .

12.5

50% (: $300 → $450) :

- : +50%($16.4B → $24.6B).
- : +50%($1,250 → $1,875).
- / : +50%( ).

(2% → 4%) :

- : +100%( ).
- TEM : 20% .
- : 8% 10% , 12% 11% .

TOS TOS . TOS $5 $15 :

- : 500 TOS = $7,500($2,500).
- : 1,000 TOS = $15,000($5,000).
- : . TEM (: 500 → 200 TOS).

12.6 (MERCURY VENUS)

MERCURY VENUS :

- **MERCURY()**: $16.4B , 50,000 , 500 .
- **MERCURY()**: $180B (12%), 800,000 , 2,500 . PAI 10,000+ TPS , .
- **MERCURY / VENUS()**: $650B , 4.5M ( ), 10,000 . DAE , . TOS $50B .

12.7

: MERCURY 5% $500M . :

- (3%) .
- (, Section **??**).
- (50%); 0 .
- : 72.
- : , (v5.0), .

: AI AI . :

- : TOS EU ; 15% .
- : EU TOS ; ; .

13

TOS . , , .

13.1

T1: ( ) : ; .
: ID , , .
:

- :
- : .
- : .

: (2016), (2020–2023), Amazon/Yelp .

T2: : .
: $> 33\%$.
:

- :
- : .
- : ID .

: , MEV .

T3: : CC/EC .
: , , .
:

- : TEM .
- : .
- : API (AWS ,) .

: LIBOR (2008–2012), (2010), .

T4: Capture : , , .
: , , .
:

- : , , .
- : .
- : .

: , , .

T5: : , , .
: , , .
:

- : TaskInstance RewardSplitter .
- : .
- : DoS.

: The DAO (2016, $50M), Parity (2017), Poly Network (2021, $600M).

13.2

1:

- **TEE** : Intel SGX/AMD SEV , . : (Spectre, Meltdown); zk-SNARK .
- : Dilithium(), Falcon( ) . : v4.0 (+ PQC)(C, PQ-2025-08).
- : BLS , t-of-n .

2: ()

- : (100 TOS / 1,000 TOS); (1-50%) . : $> \$500k$; > 30 ().
- : RepGraph 2% , .
- : (5-60) , DoS .

3:

- : 5 , ≥ 3 ($< 0.01\%$).
- : + (drand) VRF( ) , .
- : TaskInstance TLA+ , ( ) ( ) . : `github.com/tos-network/formal-verification`.

4:

- : ML (: > 10 , $> 95\%$) . (SOC) .
- : (1) $> 10\%$, (2) 1 > 5, (3) $> 20\%$ . .
- : $50k-$500k . Immunefi/HackerOne , TOS .

5:

- : 4-of-7 (1) , (2) , (3) (, Section **??**) . , 72 .
- : v5.0 ID Worldcoin/BrightID . .
- : (1) , (2) , (3) , (4) . API .

13.3

1: (0–15) SOC ; (P0: , P1: , P2:). ( ).

2: (15–60) P0 ; ; . ; .

3: (1–24) : , , . ; (,).

4: (1–7) (48). . .

5: () ; ; . ( ).

13.4

	(USD)	
(1,000)	15 () + 2 ()	(+)
(34%)	850 (TOS 5, 570k TOS)	200 ( );
(11/21)	200 () + 50 ()	500 ( );
	5 20 ( )	0 5,000 ( );
(51%)	1,200 ( )	1,000 ( );

Table 11: TOS - . 2025 10 ; .

: . : (,) . , .

14

( ), ( ) ( ) . TOS , .

14.1

(TOS) : TOS .
:

- (,) .
- (, 3) .
- (TOS $100k) /.

: ; $= \sqrt{\text{TOS}}$. : 10,000 TOS = 100; 1,000,000 TOS = 1,000(100,000). .
: (: PQC , TEM). , .

: AI ; .
:

- AGIW (, ,).
- (: AI ,).

: . ($r \geq 0.8$) 1.5× .

: ; AGIW ; .
:

- (BlockDAG , PAI).
- (, ,).

: ; > 5% 5% ( ).

: , , , AI , 7–15 .
: 3, ($\frac{1}{3}$).
:

- : (4-of-7).
- : $100k ; .
- : ; ; .
- : ; (: EU AI Act, US AI RMF).

: ; 67% .

: 5 + 3 .
:

- : ; 7 .
- : , , .
- : ; .

: 2% ; .

: ; ; ( ).
: API (, ,). .
: ; , AML/KYC , .

14.2

1: (7–30) : Commonwealth, Discord, GitHub . (, , - ,) .
: . < 20% .

2: (1–3) :

- 10,000 TOS (;).
- ( ) ( ) .
- ( ).

: , , , , ( ).

3: (7, 48) : TOS 15% ( ).
:

- /: 60% ( ).
- **(<$100k)**: ( ).
- **(>$100k)**: 67% .
- : 75% + ( ).
- : 4-of-7 (; 14).

: (`vote.tos.network`), CLI, SDK . ; .

4: (48) : 48 . : ; .
: (`Governor.py`) ; (: ,).
: (4-of-7); + .

5: (30–90) : (: TEM ,).
: ; .
: ( ).

14.3

“” . (75%) .

()

1. : , (:).
2. : , .
3. : TOS ( ); ( ) .

()

1. : // (α, β, γ) ($\beta \in [0.10, 0.15]$, $\gamma \in [0.05, 0.10]$).
2. : / 50
3. : = 20% ( ).

14.4

: : $10 .
: (1) ROI . (2) . (3) (: 1,000) ; .

: : ; .
: (1) . (2) . (3) “ ” ( ).

: : .
: (1) 4-of-7 . (2) ; 14 . (3) .

14.5

1 (v3.0–v3.5, Early MERCURY)
; . : , .

2 (v4.0, Mid-MERCURY)
, , . .

3 (v5.0, Late MERCURY)
AI ( ).

4 (VENUS+)
-AI ; .

15

TOS , AI . TOS .

15.1

		TOS
AI (NIST AI RMF, 2023)	, ,	, API,
(Howey Test)		, ,
AML/KYC (FinCEN)		; KYC
AI Act (2024)	AI ,	, , CE
GDPR	, ,	, DID,
MiCA ( )	,	CC/EC;
MAS		;
AI	, ,	RepGraph , ,
(2022)	, ,	; ( )
AI ( )	AI	(FCA)

Table 12: TOS .

15.2

(AGIW , ,) . API :

- (,).
- OAuth2 (RBAC); 7 .
- (C).

TOS Chainalysis/Elliptic . / KYC (: Circle, Paxos) . ; .

(Pedersen + Bulletproofs) . GDPR “” :

- ( ) (IPFS, Arweave).
- DID .

EU-US (2023) . TOS (SCC) (: ,).

15.3

1. : ; (: EU AI Act).
2. : (MAS, FCA) .
3. : Blockchain Association, AI Safety Alliance, IEEE P3119(AI) .
4. : (**??**) (: “OFAC ”).

16

16.1

(Themis): ; 2025 9 ; 42 347 .
(Helios): ; ; -TOS .
(Aurora): ; ; .
(Metis): ; .

16.2 (2025 3)

		(v4.0)
(TPS)	2,847 , 3,521	PAI 10,000+
	11.3s , 18.2s p95	<10s
	8 (90s)	tx 500ms
	99.7% ()	99.9% (SLA)
AGIW	14.2 min ()	<10 min
	1.8%	<2%
	45 MB/s	150 MB/s (PAI)
	128 GB	+

Table 13: TOS (Themis, 2025 9).

16.3

:

- 40% AGIW task submissions (: 10 KB–5 MB).
- 30% CC/EC swaps (AMM transactions).
- 20% DID (,).
- 10% + .

: 2 500 TPS 5,000 TPS ; , , . : `github.com/tos-network/benchmarks`.
: 10% ; ; . .

17

17.1 1:

(Fortune 500); 1,200 ; >10,000 .

250 , , . 18,000 ($4.5M , 6) . : 90.

TOS

1. : TOS 50 AI (GPT-4 ,) .
2. : 12,500 (200); = 50 CC + 5 EC.
3. : Intel SGX , + .
4. : 5 ; (3%).
5. : AGIW (, ,) .

- : $1.8M(60%).
- : 67(25%).
- : (500) 97.2% .
- : AI (, 2025).

Table 14: : ROI

	TOS	
$4.5M	$1.8M	-60%
180 ()	67	-63%
18,000	450 ()	-97.5%
$1.80	$0.72	-60%
–	12,500	–
–	14.3	–
QA	97.2% ()	
	3% ()	
	( )	

: AGIW AI "" .

17.2 2:

(23 , 5).

4.8 (MODIS, Landsat) (2015–2025). (AWS/GCP) $12M ; = $3M.

TOS

1. : 20% 2.4M CC (TEM).
2. : 300 (YOLO) ; GPU EC .
3. : / 10% ; EC .
4. : Merkle + ; TOS .

- : $2.7M (77%;).
- : 14 (24).
- : UN COP30 ; 94% .
- : + ; EC .

Table 15: : ROI

		TOS	
	$12M ()	$2.7M	-77.5%
	24	14	-42%
	4.8 PB	4.8 PB	–
TB	$2,500	$562.50	-77.5%
CC	–	2.4M	–
TEM ()	$0	$540K (20%)	
		67% (EC)	
	–	300 ()	
	–	94% ( )	
	$0	$42K (EC)	

: TEM . EC .

17.3 3:

AAA ; 1,500 ; = GMV $200M.

AI (,). : 45,000 , 6 . (EU) .

TOS

1. **NFT**: AI ; .
2. **AGIW** : (, ,).
3. : 50 TOS ; 48 ; .
4. : , , .

- : (NPS) 42 68 .
- : 2.1 (71%).
- : EU DSA TOS .
- : 120,000 TOS (12%); = $1.2M.

Table 16: : ROI

	AI	TOS	
	$3.8M	$2.6M	-32%
	45,000	2,300	-95%
()	6	2.1	-96%
NPS	42 ()	68 ()	+62%
	0% ()	100% ()	
EU DSA			
	$0	$120K ()	
	18% ()	4.2% ()	-77%
		4.3/5.0	
()	$0	50 TOS ($125)	

: . .

18 : 12

2026 1 2026 4 TOS , AGI .

18.1 2026 1: ,

:

- **DID/ v1.0**: W3C DID; ; KYC/AML .
- **AGIW v1.0**: TEE (Intel SGX, AMD SEV-SNP); ; .
- **v1.0**: , , , ; .
- **CC** : (GPU) (CC); (AMM) .
- **Paymaster** : ; .
- : / / / / `explorer.tos.network`.

: 1,000+ ; 500 /; 95%+ ; <20.

18.2 2026 2: ,

:

- **v1.0**: RepGraph ; ; API.
- **EC** : kWh (EC); ( ).
- **v1.0**: Anthropic Constitutional AI, OpenAI Moderation API, Google Perspective ; ; .
- : (:); GDPR/CCPA .
- : ; ; ; .
- **API**: TPS, , AGIW , (30) `metrics.tos.network/api`.

: 2,000+ ; 1,500 /; 80%+; <0.5%; EC $5M+.

18.3 2026 3: ,

:

- **AGIW-ZK Pilot**: AI (,) zk-SNARK ; TEE .
- **v1.0**: ; ; ; .
- **v1.0**: ; ; .
- : HTLC (CC/EC); .
- : ; KYC/AML ; CC/EC / .
- **SDK v1.0**: , , Python, JavaScript, Rust ; .

: 3,500+ ; 3,000 ; ZK 85%+; <48; TVL $10M+.

18.4 2026 4: TEM ,

:

- **TEM v1.0**: (100%, 75%, SMB 50%, 0%) TOS ; .
- : ; ; .
- : (TPS, , AGIW ,); ; .
- **3** : (: AI) ; AGIW SDK; .
- **v2.0**: TEM ; ; ; .
- : 3 (Trail of Bits, OpenZeppelin); ; .
- **PAI** : Power of AI ; AI ; BlockDAG .

: 5,000+ ; 10,000 ; <15; <2%; TEM $50M+; 5+.

18.5

(2026 4):

- : 5,000+
- : 10,000+
- AGIW : <15
- : <2%
- : >99%
- : >98%
- CC/EC : $100M+
- : 10+

:

- **ZK** : TEE ; ZK .
- : ; ; .
- : ; .
- : ; ; ; .
- : ; ; ; .

19

TOS AGI . 2025 10 .

19.1

AGIW : TEE (Intel, AMD) ; (Spectre, Meltdown).
: / zk-SNARK TEE .
: AI (,) Plonky2( ) . TEE .
: v4.5 (2026 2); v5.0 (2026 4).
: Gabizon et al. (PLONK, 2019).

(FHE) : ; .
: AGIW (: ML).
: TFHE (Zama) ; ; (: $< 10\times$).
: (2026–2027); (2028).

19.2

: ; TEM .
: + TOS ; .
: ; .
: 2026 1.

: , , .
: () .
: Gitcoin Grants ; TOS ; RepGraph .
: 2026 3 ($500k).

19.3

AI : AI ; .
: ; TOS .
: IBC( ) ; ; .
: v4.2-v4.5 (2026 4 - 2027 2).

AI AI : (, ,) .
: Anthropic Constitutional AI, OpenAI Moderation API, Google Perspective .
: ( ); < ; .
: v1 (2026 2); AI ( ).

19.4

: (2025 9 128 GB); .
: (30); .
: ; .
: v4.0 (2026 1).

Sharding for PAI Consensus : PAI 10,000 TPS .
: (: ,); 2 .
: ; AI .
: (2026–2028); (v6.0, 2029+).

19.5

- **AI** : AI ? ?
- : ?
- : -AI ?
- : (MARS/JUPITER) ?

, . : `research@tos.network`.

20

TopoSpartan TOS . ; AGI . .

A AGI : 10

. 5 , 10 50 .

A.1 MERCURY (20252075):

; “-” .

- : ; AI .
- : ; .
- : **AI** **(DAE)**; . (// +) AI .

, AI , DID, , .

, / , , .

A.2 VENUS (20752125):

AI ; .

- : **Metaworld 1.0** (: //).
- : **Cognitive Earth Union** (-AI).

AI , , .

AGI ; “” “ ” .

A.3 MARS (21252175):

-- ; .

- /// .
- LEO (ISRU +).

/ ; (kWh-).

, , .

A.4 JUPITER (21752225):

/ ; **-** .

- -- ; .
- , , .

; - ; / .

(//), .

A.5 SATURN (22252275):

; / .

- ; / ; AI .
- : “ ” .

, , - .

; .

A.6 URANUS Era (22752325):

/ ; .

- **(SFP)**: .
- , .

, , .

, .

A.7 NEPTUNE (23252375):

(, ,) .

- : /.
- // - “ ” .

/; .

; .

A.8 PLUTO Era (23752425):

, , .

- (+) .
- ; .

, , .

/ ; / .

A.9 PROXIMA Era (24252475):

; .

- : .
- .

, , .

; .

A.10 ANDROMEDA Era (24752500): -

.

- - .
- : , , .

, / , .

; .

A.11

- : → → → → .
- : → → → → .
- : “” , “” .

B AGI : (20252100)

MERCURY VENUS AGI 75 . TOS - .

B.1 I: AI (20252035)

2026 ; AI .

2028 EU **AI** AI .

2029 **I** .

2030 **AI** (, CEO).

2032 GDP 12% AI AI .

2034 - **-AI** “ ” .

B.2 II: (20352050)

2037 AGI **Aletheia-1** / (GLI $\geq$).

2038 **ID** ; AI .

2040 **AI** : // $>$ 40% AI ; .

2043 AI ; AGI .

2045 AI .

2048 AI UN ; .

B.3 III: AI (20502070)

2051 AI **(DAE)** .

2055 **TOS Metanet Singularity Graph** AI .

2058 ($\leftrightarrow$).

2061 **AI** (SolarMind) ; AI “” .

2065 : .

2068 -AI .

B.4 IV: (20702090)

2072 **1.0**: (//) .

2076 (“ ”) .

2080 : AGI .

2085 ; .

2089 , ; “ ” .

B.5 V: 10 (20902100)

2092 : AGI .

2095 , AI .

2098 : .

2100 ; .

B.6 (20252100)

- : → → → .
- : → → → .
- : , .

AGIW (PoIW) **AGI Work** (Proof-of-Intelligent-Work): , AI . AGIW .

PAI **Power of AI**: AI , (10,000+ TPS) TOS . 3 (MERCURY / VENUS) .

TEM **TOS Energy Model**: . TEM (100%), (80%) (0-50%) .

CC **Compute Credits**: . CC GPU/CPU .

EC **Energy Credits**: . EC kWh .

DID **Decentralized Identifier**: , W3C . : `did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnz` DID , .

RepGraph **Reputation Graph**: (30%), (40%), (30%), (+20%) (+10%) . 0.0 1.0.

BlockDAG 1-3 , .

DT-Settlement **Delay-Tolerant Settlement**: ( ) . (LCD) (IRC) .

HSM **Hardware Security Module**: . TOS .

TEE **Trusted Execution Environment**: (: Intel SGX, AMD SEV-SNP). TOS AI TEE .

ZK Proof **Zero-Knowledge Proof**: . TOS AI zk-SNARK (TEE).

Celestial Eras 2025-2500 10 : MERCURY (2025-2075) ANDROMEDA (2475-2500), .

AI/AGI /

API

BLS Boneh-Lynn-Shacham ( )

DAO

DAE

ECDSA

HTLC

KYC/AML /

LCD/IRC /

MEV

NIST

PQC

RPC

SOC 2 2 ( )

TPS Transactions Per Second

VC (W3C)

W3C

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.* https://bitcoin.org/bitcoin.pdf

Buterin, V. (2014). *Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform.* Ethereum White Paper. https://ethereum.org/en/whitepaper/

Lindholm, T., Yellin, F., Bracha, G., & Buckley, A. (2014). *The Java Virtual Machine Specification, Java SE 8 Edition.* Oracle America, Inc. https://docs.oracle.com/javase/specs/jvms/se8/html/

Wood, G. (2014). *Ethereum: A Secure Decentralised Generalised Transaction Ledger.* Ethereum Yellow Paper.

Bloomberg Intelligence (2024, June). *Generative AI Market Outlook.* Bloomberg L.P.

IDC (2024, April). *Worldwide Autonomous Agent Platforms Forecast, 2024–2028.* International Data Corporation.

McKinsey Global Institute (2023, June). *The economic potential of generative AI: The next productivity frontier.* McKinsey & Company.

Gartner (2024, October). *Top Strategic Technology Trends 2025.* Gartner, Inc.

Accenture (2024, August). *Autonomous Agents Are Reshaping Enterprise Automation.* Accenture Technology Vision 2024.

IBM Research (2024, May). *AI Safety Metrics for Enterprise Deployment.* IBM Corporation.

The Economist (2024, July 13). Meet the agentic future. https://www.economist.com/

Forbes (2024, August 19). Why Enterprises Need Verifiable AI Agents. *Forbes Technology Council.*

Ball, M., Rosen, A., Sabin, M., & Vasudevan, P. N. (2017). *Proofs of Useful Work.* IACR Cryptology ePrint Archive, Report 2017/203. https://eprint.iacr.org/2017/203

Gabizon, A., Williamson, Z. J., & Ciobotaru, O. (2019). *PLONK: Permutations over Lagrange-bases for Oecumenical Noninteractive arguments of Knowledge.* IACR Cryptology ePrint Archive, Report 2019/953.

Sporny, M., Longley, D., & Chadwick, D. (2022). *Verifiable Credentials Data Model v1.1.* W3C Recommendation. https://www.w3.org/TR/vc-data-model/

Reed, D., Sporny, M., Longley, D., Allen, C., Grant, R., & Sabadello, M. (2022). *Decentralized Identifiers (DIDs) v1.0.* W3C Recommendation. https://www.w3.org/TR/did-core/

ISO/IEC 20022 (2013). *Financial services – Universal financial industry message scheme*. International Organization for Standardization.

NIST (2022). *Post-Quantum Cryptography: Selected Algorithms 2022*. NIST FIPS 203-205. National Institute of Standards and Technology.

W3C (2018). *Verifiable Credentials Data Model 1.0*. W3C Recommendation. World Wide Web Consortium.